

Initial Project Planning Template

Date	06 July 2026
Team ID	XXXXXX
Project Name	Voice Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation (4 Marks)

Use the below template to create a product backlog and sprint schedule

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Concept & Audio Input	USN-1	As a user, I can select a concept from a predefined list to be assessed on.	2	High	Hasini	10 Feb 2026	21 Feb 2026
Sprint-1		USN-2	As a user, I can upload or record an audio explanation of the selected concept.	3	High	Hasini	10 Feb 2026	21 Feb 2026
Sprint-1		USN-3	As a user, I can use a bundled sample audio file to try the tool without recording my own.	1	Medium	Hasini	10 Feb 2026	21 Feb 2026
Sprint-1	Speech Transcription	USN-4	As a backend service, I can transcribe uploaded audio to text using OpenAI Whisper.	3	High	Jagadesh	10 Feb 2026	21 Feb 2026
Sprint-1		USN-5	As a backend service, I can resample and preprocess audio to 16kHz mono before transcription.	2	High	Dada Khalandar	10 Feb 2026	21 Feb 2026
Sprint-2	Semantic Scoring	USN-6	As a backend service, I can compute semantic similarity between the transcript and the reference concept using Sentence-BERT.	3	High	Karthik	24 Feb 2026	07 Mar 2026
Sprint-2		USN-7	As a backend service, I can identify missing key terms (gap analysis) between the transcript and the reference concept.	2	Medium	Karthik	24 Feb 2026	07 Mar 2026

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Sprint-2	Audio Fluency Analysis	USN-8	As a backend service, I can detect pauses and silence in the audio using Librosa.	2	High	Dada Khalandar	24 Feb 2026	07 Mar 2026
Sprint-2		USN-9	As a backend service, I can count filler words ("um", "like", etc.) in the transcript.	2	Medium	Dada Khalandar	24 Feb 2026	07 Mar 2026
Sprint-2		USN-10	As a backend service, I can compute pitch, RMS energy, and speaking rate (WPM) from the audio.	3	Medium	Dada Khalandar	24 Feb 2026	07 Mar 2026
Sprint-2	Scoring & Feedback	USN-11	As a backend service, I can combine semantic and fluency scores into a single weighted score.	2	High	Karthik	24 Feb 2026	07 Mar 2026
Sprint-2		USN-12	As a user, I receive a written feedback paragraph explaining my score.	2	Medium	Karthik	24 Feb 2026	07 Mar 2026
Sprint-3	Backend API	USN-13	As a frontend client, I can call POST /api/analyse to submit audio and receive scoring results as JSON.	3	High	Jagadesh	10 Mar 2026	21 Mar 2026
Sprint-3		USN-14	As a frontend client, I can call GET /api/report/{session_id} to download a PDF report.	2	Medium	Jagadesh	10 Mar 2026	21 Mar 2026
Sprint-3	Results Visualization	USN-15	As a user, I can view my results with waveform, radar, and donut charts.	3	High	Hasini	10 Mar 2026	21 Mar 2026
Sprint-3		USN-16	As a user, I can see the highlighted keyword gaps in my transcript.	2	Low	Hasini	10 Mar 2026	21 Mar 2026
Sprint-3	PDF Reporting	USN-17	As a user, I can download a PDF report summarising my transcript, scores, and charts.	3	Medium	Karthik	10 Mar 2026	21 Mar 2026
Sprint-3	Deployment	USN-18	As a developer, I can run the frontend and API through a Node/Express proxy server locally.	2	Medium	Jagadesh	10 Mar 2026	21 Mar 2026